

Pombiteer

- **Diggante waxtu:** 10 simili
- **Environment:** benn GPU (≈16 GB VRAM), internet amul
- **Dayob solution:** `solution.ipynb` ≤ 1 MB
- **Stockage:** 5 GB

Liggéey bi

Sa xarit digal la ngeen fo benn po bu ñuy seet baat. Moom, muy àttekat bi, dina tànn benn baat bu làqu ci vocabulary bu ñu saxal, te war nga ko gis ci lu éppul 30 tour. Ci bépp tour, àttekat bi dina méngale ñaari baat, ba wax ban moo ci maana gën a jege baat bi làqu. Po bu nekk dafay tàmbaalee ci paire bi ñu saxal `lamp` vs `potato`, ndaxte ñoom ñaar ñoo bakk ci mbir yi sa xarit gën a bëgg. Sa program bi dina joxe benn baat bu bees. Baat bi raw ci méngale bi, dees koy denc te méngale ko ak sa jooxe bi ci topp. Dinga daan po bi saa si nga joxee baat bi làqu ci anam wu dèppoo. Dèppale bi case-insensitive la. Bépp baat boo joxe war na nekk ci `dataset/vocabulary.json`.

Am na misaal bu mat sèkk ci `solution.ipynb`, ànd ak protocol ak charger donnees yi. Mën nga soppi class `PublicEmbeddingPlayer`. Dees na initialise sa program bi benn yoon, mu fo po yépp ci benn doxal; protocol bi dafay sos `PublicEmbeddingPlayer` bu bees ci njàlbéenug bépp po.

Àttekat bi

Sa program bi dafay yónnee Àttekat bi benn objet JSON, Àttekat bi tontu ak benn objet JSON.

Misaal bu baax, te ñu wone baat bi làqu ngir rekk leeral protocol bi:

```
Hidden word: shovel          Fixed opening: lamp vs potato

<- {"turn": 1, "winner_word": "potato", "verdict": "second", "word1": "lamp", "word2": "potato"}
-> {"new_word": "rock"}
<- {"turn": 2, "winner_word": "rock", "verdict": "second", "word1": "potato", "word2": "rock"}
-> {"new_word": "hammer"}
<- {"turn": 3, "winner_word": "hammer", "verdict": "second", "word1": "rock", "word2": "hammer"}
-> {"new_word": "shovel"}
-> {"status": "win"}

<- matches: game won
```

Ñu ngi index tour yi dale ko ci 1 ba 30.

Tàñneefi verdict yi ñooy `first`, maanaam `word1` moo gën a jege, `second`, maanaam `word2` moo gën a jege, walla `same`, maanaam baat yépp ñoo yem ci ni ñu jege baat bi làqu.

`winner_word` mooy baat bi ñuy denc ngir méngale gi ci topp. Su verdict bi dee `same`, baat bu njëkk bi dafay des.

Dataset

Xajalé yépp dañuy bakk yii:

- `dataset/vocabulary.json` — 1602 baat yu bakkul te ñépp miniscule lañu. Baat bi làqu dafay nekk saa su nekk benn ci yii.
- `dataset/public_embeddings.npy` — `float32`, shape (1602, 2560). Row `i` dafay méngoo ak baat `i` ci vocabulaire bi. Yii ay embedding `public` lañu; àttekat bi dafay jëfandikoo representation bu wuute te private.

Xajalé yi ay mbooloom baat yu làqu lañu:

Xajalé	Baat yi	Tontu yi	Jëfandikoo ko ngir
<code>test_public</code>	120	<code>dataset/test_public.json</code>	doxal sa solution te xayma ko
<code>test_leaderboard_a</code>	120	làqu	classement buy dox
<code>test_leaderboard_b</code>	120	làqu	classement bu mujj

Amul Xajale `train` — dara du nu ko jele ci ligne yu am label.

Model yi ñu joxe

Ñaari pretrained embedding model dañu ànd ak liggéey bi te mën nañu leen jëfandikoo:

```
models/Qwen/Qwen3-Embedding-0.6B
[Qwen Embedding model card] (https://yastatic.net/s3/contest/ioai/3/01_Qwen_Qwen3-Embedding-0.6B_MODEL_CARD.html)
models/BAAI/bge-m3
[bge-m3 model card] (https://yastatic.net/s3/contest/ioai/3/02_BAAI_bge-m3_MODEL_CARD.html)
```

Dees na leen ñaar yépp charger ci seen chemin local; Hugging Face hub id bu mel ni "BAAI/bge-m3" dafay déclencher telechargement te du antu, ndax àtte bi amul internet. Bépp dossier am na `example.py` bu mën a dox, biy wone wote bu amul internet.

Librarie yi am: `numpy`, `torch`, `sentence-transformers`. Internet amul, telecharment amul, te yenen package amul.

Génne gi

Dara. Liggéey bu interactive la: sa solution du bind fichier tontou; dafay jukkoo ak àttekat bi jaarale ci `stdin/stdout`, ni ñu ko tëral ci kaw.

Jooxe Score

Po bu ñu gis ci tour t dafay am score $1.0 - 0.02 \times \max(0, t - 10)$; po bu ñu saafaraagul ci biir 30 tour dafay am score 0. Kon tour 1–10 yi dañuy am score 1.00, tour 20 dafay am score 0.80, tour 30 dafay am score 0.60.

Score sa liggéey mooy moyenne score po yi $\times 100$, diggante 0.00 ak 100.00.

Diggante 10 simili bi benn budget la buy mboole tambali, preparation ak po yépp 120 ci test set bi.

Naka lañuy yónnee

1. Ubbi `solution.ipynb`, soppi `PublicEmbeddingPlayer`, te doxal celluless yépp ngir wóor ne dafay dox.
2. Soo bëggee, seetal ko ci local: `python local_test.py solution.ipynb --limit 5`. Àttekat bu local bi dafay jëfandikoo embedding yu *public*, kon score bi ab tegtal rekk la.
3. Sauvegarder `solution.ipynb`.
4. Ubbi Git tab bi ci sidebar bu càmmooñu JupyterLab.
5. Def **Stage** `solution.ipynb` (natal + bi ci wetam).
6. Bind commit message, ba noppi bës ci **Commit**.
7. Bës ci niir bi ànd ak fitt buy jëm kaw ngir push.
8. Dellu ci xëtum Contest bii, te bës ci **Submit**, commit message bi war a méngoo ak bi nga joxe.

Yónne benn fichier rekk, turam di `solution.ipynb`, buy mboole bépp preparation ak inference bu war.